MFAD Unit 2 FAQ

Orthogonality & Projections

1. Let $W = \{(a, b, b) \mid a, b \text{ are real}\}$ and let $v = (3, 2, 6)$. (i) Find an orthonormal basis for W. (ii) Find the projection of v onto W, say v_1. (iii) Decompose v into a sum of two vectors $v_1 + v_2$ where v_2 is projection of v onto $W^{\perp}$. **[ESA Jan - May 2024]**

2. What point on the plane $x + y - z = 0$ is closest to $b = (2, 1, 0)$? **[ESA Dec 2024]**

3. What matrix P projects every point in $\mathbb{R}^3$ onto the line of intersection of the planes $x+y+z = 0$ and $x - z = 0$? **[ESA Jan - May 2024]**

4. Project $b = (1, 0, 0)$ onto the lines through $a_1 = (-1, 2, 2)$, $a_2 = (2, 2, -1)$ and $a_3 = (2, -1, 2)$. Add the three projections to get the sum b. Also find the corresponding projection matrices P_1, P_2 and P_3. **[ESA July 2023]**

5. Find the point on the plane $2x + 5y + z = 0$ that is nearest to $(2, 0, 1)$. **[ESA July 2023]**

6. If W is a subspace spanned by the orthogonal vectors $(2, 5, -1)$ and $(-2, 1, 1)$ find the point in W that is closest to $(1, 2, 3)$. **[ESA July 2023]**

Least Squares & Gram-Schmidt

1. Find $\|E\|^2 = \|Ax - b\|^2$ and solve the normal equations $A^T A\hat{x} = A^T b$. Find the solution and the projection $p = A\hat{x}$ (Use Least squares method) Given:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad x = \begin{pmatrix} u \\ v \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}$$

   [ESA Jan - May 2024]

2. Fit a plane $y = C + Dt + Ez$ to the four points by the method of least squares: $y = 1$ at $z = 1$, $t = 1$; $y = -2$ at $z = 0$, $t = 1$; $y = 1$ at $z = 1$, $t = 0$; and $y = 0$ at $z = 0$, $t = 0$. **[ESA Dec 2023]**

3. An ice-cream vendor records the number of hours of sun shine (x) versus the number of ice-creams sold in an hour (y) at his shop from Monday to Friday and found the following data: $x : 2, 3, 5, 7, 9$ and $y : 4, 5, 7, 10, 15$. Find the best values of m and c that suit the equation $y = mx + c$. **[ESA July 2023]**

4. Use the Gram - Schmidt process to find a set of three orthonormal vectors from the independent vectors $a_1 = (1, 1, 1, 1)$, $a_2 = (1, 1, 2, 4)$ and $a_3 = (1, 2, -4, -3)$. **[ESA Dec 2024]**

5. Apply Gram Schmidt process to orthogonalize the basis $\{(1, -1, 1), (1, 0, 1), (1, 1, 2)\}$. **[ESA Jan - May 2024]**

6. Use the Gram - Schmidt process to find a set of orthonormal vectors from the independent vectors $a_1 = (1, -1, 0)$, $a_2 = (0, 1, -1)$ and $a_3 = (1, 0, -1)$. **[ESA Jan - May 2024]**

7. Find an orthonormal set q_1, q_2, q_3 for which q_1 and q_2 span the column space of $A = \begin{pmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{pmatrix}$. Which fundamental subspace contains q_3? **[ESA Dec 2023]**

8. Use the Gram - Schmidt process to find a set of orthonormal vectors from the independent vectors $a_1 = (1, 1, 1)$, $a_2 = (0, 1, 1)$ and $a_3 = (0, 0, 1)$. Also find the $A = QR$ factorization. **[ESA Dec 2023]**

Eigenvalues & Diagonalization

1. Find the eigen values and the eigen vectors of:

$$\begin{pmatrix} 2 & 1 & 2 \\ 2 & 2 & -2 \\ 3 & 1 & 1 \end{pmatrix}$$

[ESA Dec 2024]

2. Find the eigenvalues and the corresponding eigenvectors of $\begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}$. **[ESA July 2023]**

3. Check if the following symmetric matrix $A = \begin{pmatrix} 7 & 4 & 4 \\ 4 & 5 & 0 \\ 4 & 0 & 9 \end{pmatrix}$ is orthogonally diagonalizable. If so, orthogonally diagonalize it as $A = S\Lambda S^{-1} = Q\Lambda Q^{-1} = Q\Lambda Q^T$ where Q is an orthogonal matrix. **[ESA Jan - May 2024]**

4. Find the matrices S and S^{-1} to diagonalize $A = \begin{pmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}$. What are limits of Δ^k and $S\Delta^k S^{-1}$ as $k \to \infty$? **[ESA July 2023]**

5. Consider the quadratic form $Q = 2x_1x_2 + 2x_1x_3 + 2x_2x_3$. Write down the matrix 'A' of the quadratic form and diagonalise 'A' to get the sum of squares form of Q. **[ESA Dec 2024]**

6. Find the matrix A whose eigen values are 2 and 5, and whose eigen vectors are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ respectively using $S\Delta S^{-1}$. **[ESA July 2023]**

7. Diagonalize the matrix $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{pmatrix}$. **[ESA Jan - May 2024]**

Definiteness, SVD & Covariance

1. For what values of 'a' will the matrix $\begin{pmatrix} a & -1 & -1 \\ -1 & a & -1 \\ -1 & -1 & a \end{pmatrix}$ become positive definite? **[ESA Dec 2024]**

2. For which a are the matrix A have all $\lambda > 0$ and are therefore positive definite. $A = \begin{pmatrix} a & 2 & 2 \\ 2 & a & 2 \\ 2 & 2 & a \end{pmatrix}$. **[ESA July 2023]**

3. Test if $A^T A$ is positive definite or positive semi-definite given $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{pmatrix}$. **[ESA Jan - May 2024]**

4. Verify for the positive definiteness of the matrix $\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$. **[ESA Jan - May 2024]**

5. The following table lists the weights and heights of 5 boys:

Boy	**1**	**2**	**3**	**4**	**5**
Weight (lbs)	120	125	125	135	145
Height (in)	61	60	64	68	72

   Find the covariance matrix for the above data. **[ESA July 2023]**

6. Find SVD of the matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$. **[ESA Jan - May 2024]**

7. Find the singular value decomposition of the matrix $A = \begin{pmatrix} -4 & -7 \\ 1 & 4 \end{pmatrix}$. **[ESA Dec 2024]**

8. Find SVD of the following matrix $A = \begin{pmatrix} 4 & 4 \\ -3 & 3 \end{pmatrix}$. **[ESA July 2023]**